

H2-Newton, Ariza Ikhfoula IT-2002 703

$$f(x,y) = 2x^2 + e^y - 5y \quad x^{(0)} = (3, 0)$$

1. gradient: $\nabla f(x,y) = (4x, e^y - 5)$

2. Hessian: $H(x,y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & e^y \end{pmatrix}$

3. minimizer: $\nabla f(x^*, y^*) = 0$

$$\begin{cases} 4x = 0 & \rightarrow x^* = 0 \\ e^y - 5 = 0 & \rightarrow y^* = \ln 5 \approx 1.6094 \end{cases}$$

$$(x^*, y^*) \Rightarrow (0, \ln 5) \approx (0, 1.6094)$$

4. For every real y , $e^y > 0$,

$$H(x,y) = \begin{pmatrix} 4 & 0 \\ 0 & e^y \end{pmatrix} \text{ is PD,}$$

f is strictly convex, $(0, \ln 5)$ - unique global minimizer

5. min value: $f^* = 2(0)^2 + e^{\ln 5} - 5 \ln 5$

$$f^* \approx -3.0442$$

Newton Steps:

$$H(x^{(k)}) p^{(k)} = -\nabla f(x^{(k)})$$

$$x^{(k+1)} = x^{(k)} + p^{(k)}$$

$$\begin{pmatrix} 4 & 0 \\ 0 & e^y \end{pmatrix} \begin{pmatrix} p_x \\ p_y \end{pmatrix} = -\begin{pmatrix} 4x \\ e^y - 5 \end{pmatrix} \quad \begin{aligned} 4p_x &= -4x \rightarrow p_x = -x \\ e^y p_y &= -(e^y - 5) \end{aligned}$$

$$p_y = \frac{5 - e^y}{e^y} = 5e^{-y} - 1$$

$$p = (-x, 5e^{-y} - 1)$$

$k=0$: $x^{(0)} = (3, 0)$

$$\nabla f(3, 0) = \begin{pmatrix} 4 \cdot 3 \\ e^0 - 5 \end{pmatrix} = \begin{pmatrix} 12 \\ -4 \end{pmatrix}$$

$$f_0 = \begin{pmatrix} 12 \\ -4 \end{pmatrix}$$

$$H(3, 0) = \begin{pmatrix} 4 & 0 \\ 0 & e^0 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$$

$$H_0 p_0 = -g_0$$

$$\begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} -12 \\ 4 \end{pmatrix}$$

$$\begin{aligned} 4p_x &= -12 \\ p_x &= -3 \end{aligned}$$

$$p_y = 4$$

$$p_0 = (-3, 4)$$

$$X^{(1)} = X^{(0)} + p_0$$

$$X^{(1)} = (3, 0) + (-3, 4) = (0, 4) \quad f(3, 0) = 2 \cdot (3)^2 + e^0 - 5(0) = 19.0000$$

Descent direction: $\langle \nabla f, p \rangle = g_x p_x + g_y p_y$
 $= 12(-3) + (-4)(4) = -52.0000$
 $-52 < 0$, descent direction

error in y : true min $\text{los } y^* = 1.5 \approx 1.6084$
 $|y^{(0)} - y^*| = |0 - 1.6084|$

$$k=1: X^{(1)} = (0, 4)$$

$$\nabla f(0, 4) = \begin{pmatrix} 0 \\ 48.5882 \end{pmatrix} \quad H(0, 4) = \begin{pmatrix} 4 & 0 \\ 0 & 54.5882 \end{pmatrix}$$

$$H_1 p_1 = -g_1 \rightarrow \begin{pmatrix} 4 & 0 \\ 0 & 54.5882 \end{pmatrix} \begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} 0 \\ -48.5882 \end{pmatrix}$$

$$p_x = 0 \quad 54.5882 p_y = -48.5882$$

$$p_1 = (0, -0.8984)$$

$$p_y \approx -0.8984$$

$$X^{(2)} = X^{(1)} + p_1 = (0, 4) + (0, -0.8984) \approx (0, 3.0916)$$

$$f(0, 4) = 2 \cdot 0^2 + e^4 - 5(4) = 34.5882$$

$$34.5882 > 19$$

descent direction: $\langle g, p_1 \rangle = 0(0) + 48.5882(-0.8984)$

$$\langle g, p_1 \rangle \approx -43.5056, p_1 \text{ is a descent direction}$$

error in y : $|y^{(1)} - y^*| = |4 - 1.5| = |4 - 1.6084| = 2.3906$

$$k=2: X^{(2)} \approx (0, 3.0916)$$

$$f(X^{(2)}) = 2(0)^2 + e^{3.0916} - 5(3.0916)$$

$$f(X^{(2)}) \approx 6.5539$$

error in y : $|y^{(2)} - y^*| = |3.0916 - 1.6084| \approx 1.4824$

Results table

k	$x^{(k)}$	$f(x^{(k)})$	$\langle \nabla f, p \rangle$	$ y^{(k)} - y^* $
0	(3, 0)	18.0000	-52.0000	1.6094
1	(0, 4)	34.5882	-45.0560	2.3806
2	(0, 3.0816)	6.5539	—	1.4821

1) Is f monotone along Newton iterations?

$$f(x^{(k+1)}) \leq f(x^{(k)})$$

$$18 < 34.5882 > 6.5539$$

No, the function values are not monotonically decreasing

A full Newton step with $\lambda = 1$ may be too large. A step direction derivative only guarantees a decrease for positive steps, not a full step.

At $k=0$, Newton moves y from 0 to 4, overshooting $y^* = 0.5 \times 1.6094$

2) would it ~~drop~~ Newton accept $\lambda = 1$ at $k=0$?

$$\text{Armijo: } f(x^{(0)} + \lambda p_0) \leq f(x^{(0)}) + \lambda c \langle \nabla f_0, p_0 \rangle$$

$$c = 10^{-4} = 0.0001 \quad \lambda = 1$$

$$18 + (0.0001)(1)(-52) = 18.9848$$

$$34.5882 \leq 18.9848, \quad \lambda = 1 \text{ rejected}$$

3) quadratic-convergence ratio

$$R = \frac{|y^{(2)} - y^*|}{|y^{(1)} - y^*|^2} = \frac{1.4821}{(2.3806)^2} \approx 5.7156$$

$$R \approx 0.2589$$

Quadratic convergence after Newton's method is close to nil,

the error satisfies: $e_{k+1} \leq C e_k^2$. Error can become

much smaller with each iteration.

Answer